

Open Science in Practice: Collaborative Research with Git



Niklas Edvall, PhD
niklas.edvall@ki.se

K8F6106
HT2026



Andreas Gerhardsson, PhD
andreas.Gerhardsson@ki.se

Why git?

Intro and context

Niklas Edvall

Andreas Gerhardsson 08:55 [Translate](#)



Ska plocka fram lite siffror till en metanalys... hade behövt vår kurs för 8 år sedan

-  R_full_mlm.R
-  R_full_mlm2.R 
-  R_full_mlm2old_2018_2.R 
-  R_full_mlm2old_2018_corr.R 
-  R_full_mlm2old_2018.R 
-  R_full_mlm2old.R 
-  R_full_mlm201710.R 
-  R_full_nback_analysis.R 

08:55

Hahahaaa

Den där bilden kommer komma med på en slide



Intended learning outcomes

- **In this course you will learn how to:**
- Organize a script-based analysis pipeline.
- Use the version control software *git* to keep track of changes.
- Publish your analysis on the platform *github*.
- Collaboratively incorporate updates to your work.

- **After completing the course, the participants are expected to be able to:**
- Organize and document scripts and code in an easy to navigate structure.
- Use version control (git) to make secure changes and create alternative pipelines.
- Use a shared code repository to facilitate easy collaborations with others.
- Responsibly implement AI agents for assistance with coding and documentation.
- Publish their analysis methods and data in a reproducible format.

Requirements for passing course

- Active participation
- Completing a mini-project
- Present your project at seminar (23d Oct)

Requirements for mini-project

- Write a pipeline to analyze a dataset
- Minimum analysis is one figure + one "result"
- Data goes in → Results (report) comes out
- In a public repo on GitHub
- Plausible commit history
- At least one (new) branch + merge
- Well organized – no monoliths
- Well documented (README) and commented

Wiki and schedule

 **Open Science in Practice: Collaborative Research with Git**



 GitHub

HomeScheduleLecturesExercisesResources

Schedule

Course Schedule HT 2026

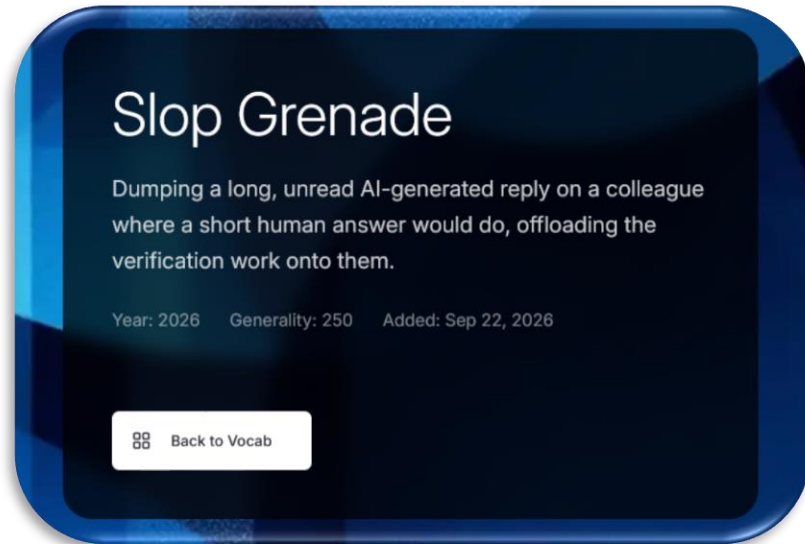
Make sure you have followed the [setup guide](#) before course start.

Download schedule: [ICS](#)

Day	Time	Activity	Material	Location
Monday 12/10				311
	09:00	Lecture	Why Git? - Intro and context	
	12:00	Lunch	Lunch places around campus	
	13:00	Lecture	Git basics	
		Exercise	Workshop #1 - Git basics	
	14:00	Lecture	Git remote	
		Exercise	Workshop #2 - Git remote	
	17:00	End of day		
Tuesday 13/10				215
	09:00	Lecture	Working in an IDE: Positron	
	10:00	Lecture	Markdown and Quarto	

Guidelines for use of AI

- Generally encouraged
- No slop grenades



Varun Anand

2w

We just instituted an official AI Writing Policy at [Clay](#). Massive thanks to [Sophie Alpert](#) on our engineering team who wrote this. Originally this was just for eng, but other teams found it so helpful we expanded it company-wide. Here are the four guiding principles:

1. You must stand behind every idea and sentence

It is your responsibility to make sure that the entire document is representative of your own thoughts before you share it.

2. Writing is thinking

Spending time on the writing process teaches you more about your topic. If you circumvent this process, you will walk away with a poorer understanding of the subject matter.

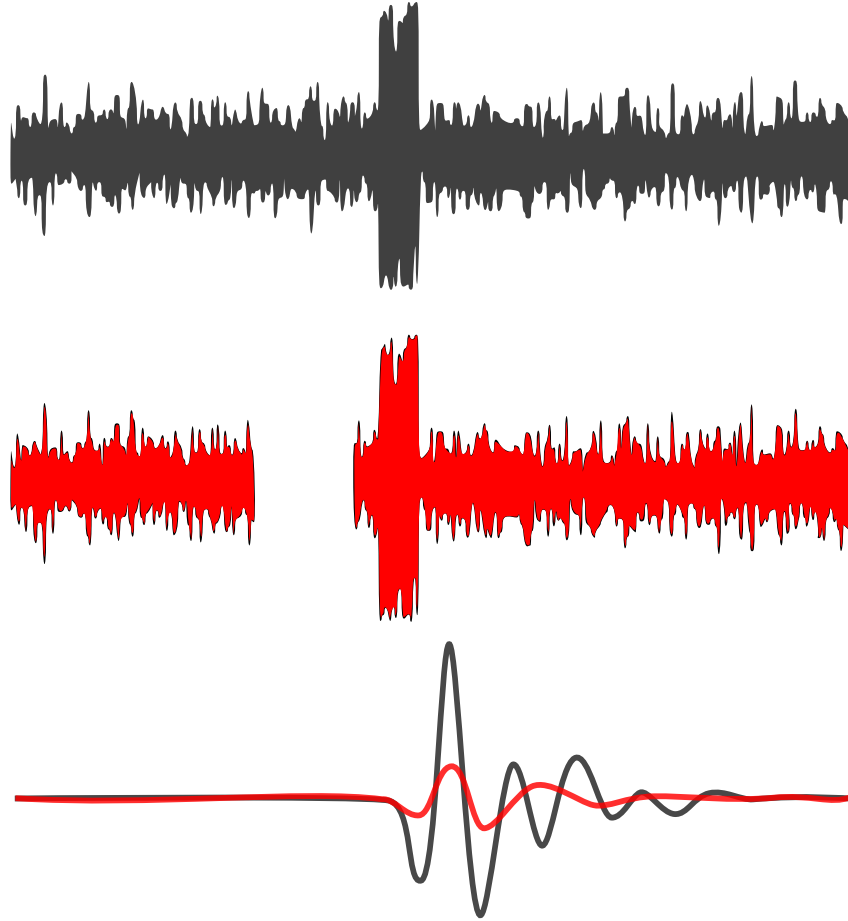
3. More time should be spent writing a document than consuming it

If you generate a document from a short prompt then ask your readers to go through the longer output, you are disrespecting their time. They can talk to ChatGPT themselves if they want to.

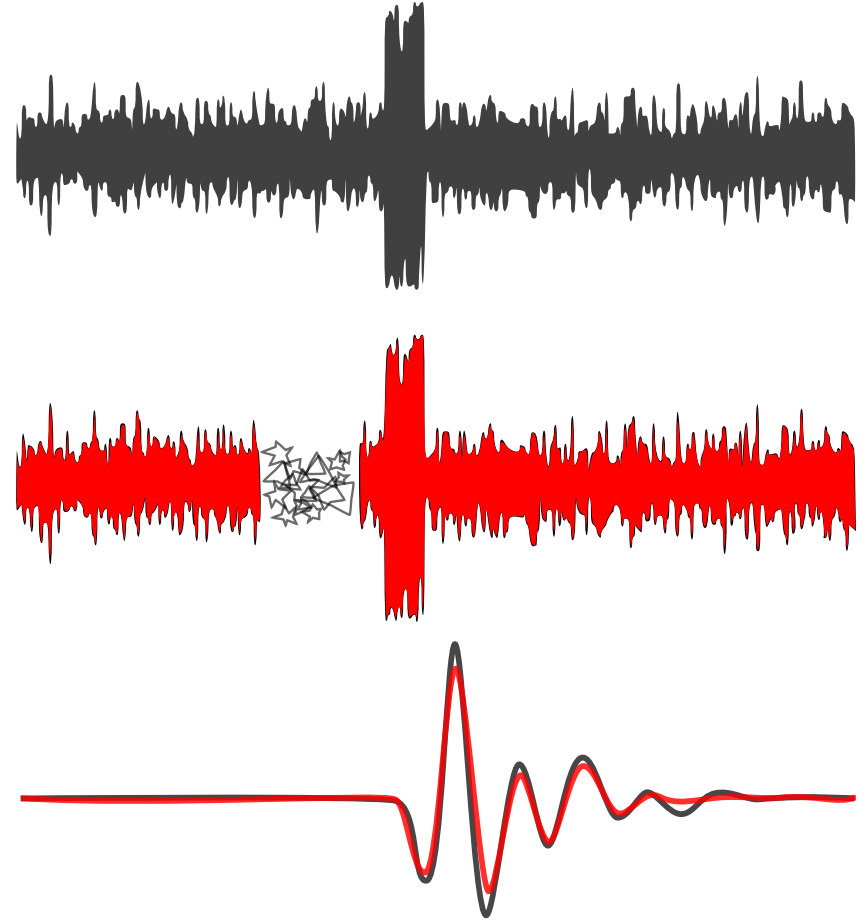
4. Longer is not better

AI makes it easy to generate long docs, and it loves padding them with sentences that say nothing. If you're producing docs from a short prompt, consider just sharing the prompt.

No Tinnitus



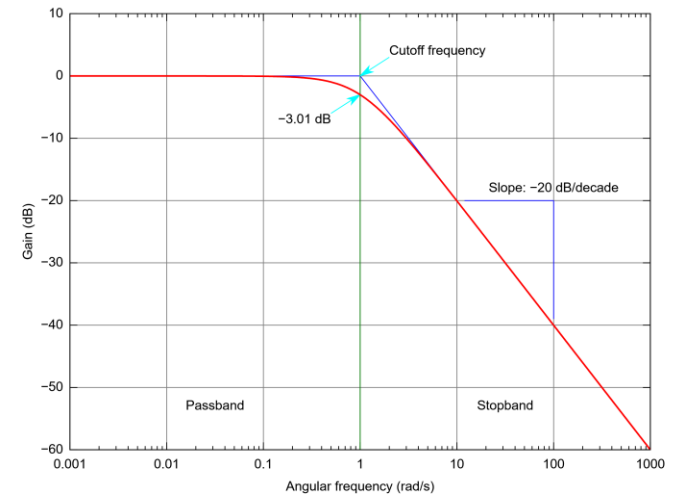
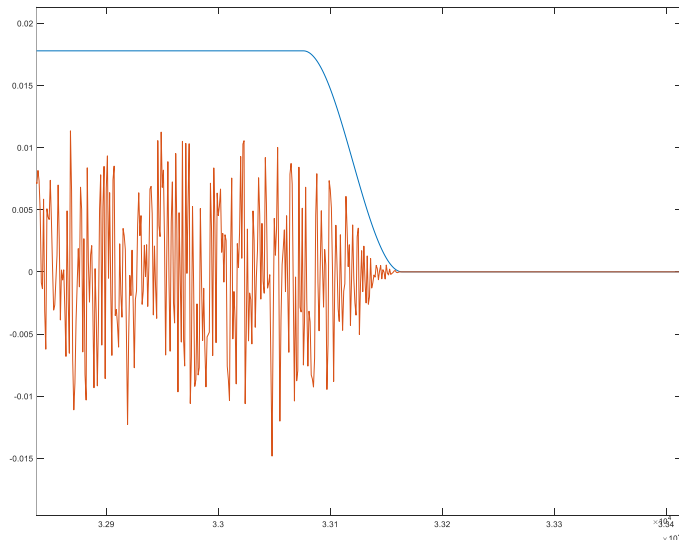
Tinnitus



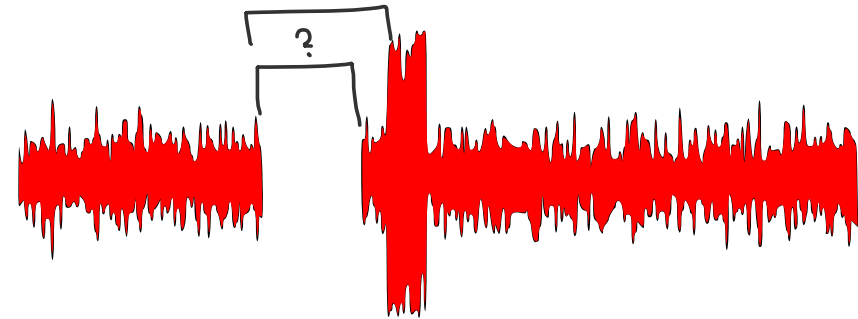
Reproducibility

Method

The stimulation paradigm consisted of white noise with a sample rate of 44.1 kHz presented bilaterally at level calibrated to 65 dBA. The background was lowpass filtered from 16 kHz with a slope of 24 dB per octave. At random intervals of between 3 and 5 seconds a silent gap with duration 50 ms was injected in the background and followed 120 ms later by a startle pulse at 95 dBA.



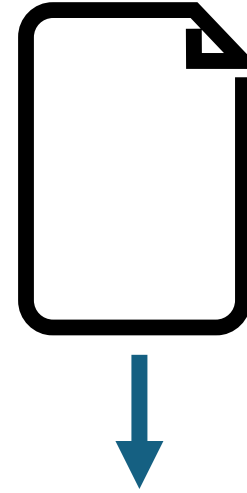
- Distribution?
- Mono channel?
- Fast? Impulse?
- What filter?
- Cutoff definition?
- Step size?
- Rise/fall time?
- From on- or offset of silent gap?
- Startle duration?



Reproducibility

Method

The stimulation paradigm consisted of white noise with a sample rate of 44.1 kHz presented bilaterally at level calibrated to 65 dBA. The background was lowpass filtered from 16 kHz with a slope of 24 dB per octave. At random intervals of between 3 and 5 seconds a silent gap with duration 50 ms was injected in the background and followed 120 ms later by a startle pulse at 95 dBA.



```
1 uni.brus <- runif(n = 44100, min = -1, max = 1)
2
3 mod <- c(rep(1,21829), rep(0,441), rep(1,21830))
4
5 final_noise <- uni.brus*mod
```

I was inspired more than 15 years ago by John Claerbout, an earth scientist at Stanford, to begin practicing reproducible computational science. See [Claerbout and Karrenbach \(1992\)](#). He pointed out to me, in a way paraphrased in Buckheit and Donoho (1995): “an article about computational result is advertising, not scholarship. The actual scholarship is the full software environment, code and data, that produced the result.” This struck me as getting to the heart of a central problem in modern scientific publication. Most of the work in a modern research project is hidden in computational scripts that go to produce the reported results. If these scripts are not brought out into the open, no one really knows what was done in a certain

JOURNAL ARTICLE

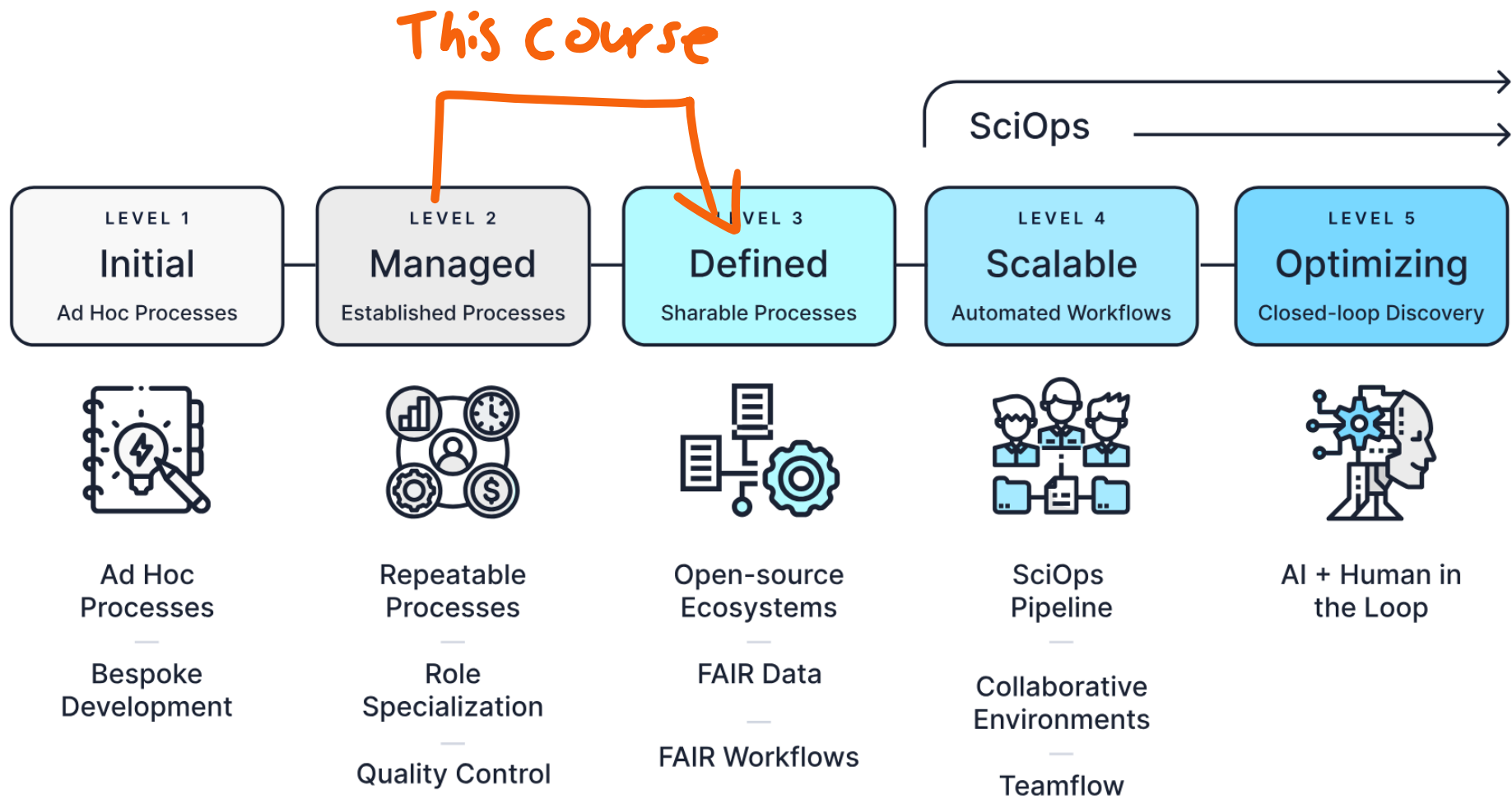
An invitation to reproducible computational research



David L. Donoho

Biostatistics, Volume 11, Issue 3, July 2010, Pages 385–388, <https://doi.org/10.1093/biostatistics/kxq028>

Published: 01 July 2010



[nature](#) > [articles](#) > articleArticle | [Open access](#) | Published: 19 May 2026

A multi-agent system for automating scientific discovery

[Ali E. Ghareeb](#), [Benjamin Chang](#), [Ludovico Mitchener](#), [Angela Yiu](#), [Caralyn J. Szostkiewicz](#), [Dmytro Shved](#),
[Gavin J. Gyimesi](#), [Jon M. Laurent](#), [Samantha M. Wright](#), [Muhammed T. Razzak](#), [Andrew D. White](#) , [Silvia C.](#)
[Finnemann](#), [Michaela M. Hinks](#)  & [Samuel G. Rodrigues](#) 

[Nature](#) **655**, 497–505 (2026) | [Cite this article](#)

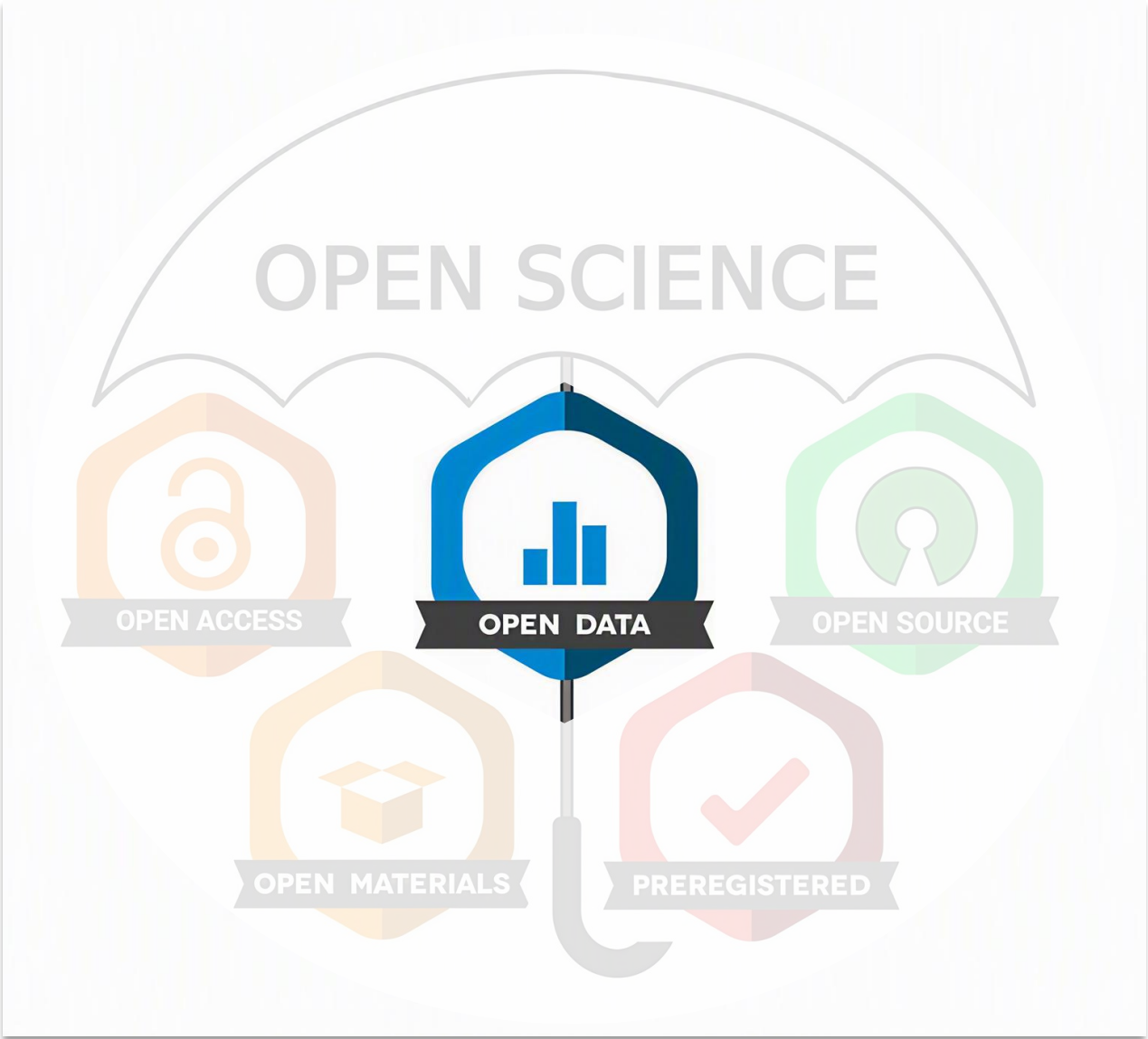
[Save article](#)

234k Accesses | **60** Citations | **519** Altmetric | [Metrics](#)

Abstract

Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages^{1,2,3}. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis

OPEN SCIENCE



The infographic features a large, light gray umbrella shape that frames the central content. At the top, the words 'OPEN SCIENCE' are written in a bold, sans-serif font. Below this, five hexagonal icons are arranged in a circle. Each icon is a different color and contains a specific symbol: an orange padlock for 'OPEN ACCESS', a blue bar chart for 'OPEN DATA', a green keyhole for 'OPEN SOURCE', a yellow box for 'OPEN MATERIALS', and a pink checkmark for 'PREREGISTERED'. Each icon is placed on top of a gray banner that contains the corresponding text. The background of the entire image is a light gray with a subtle pattern of vertical lines in white, yellow, and blue.

OPEN ACCESS

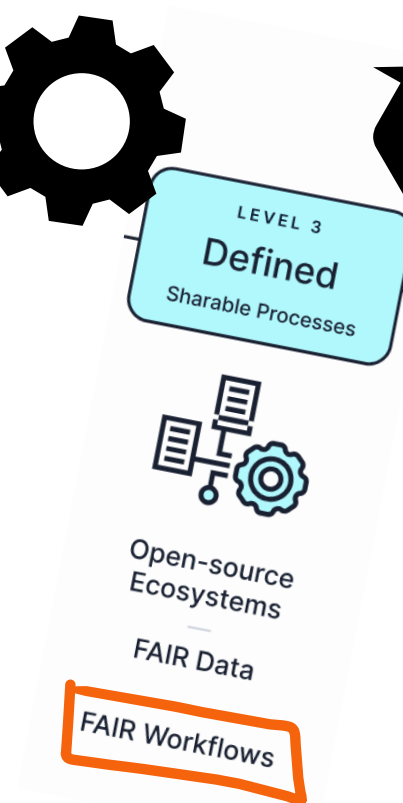
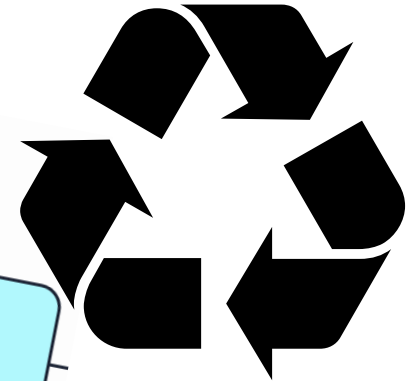
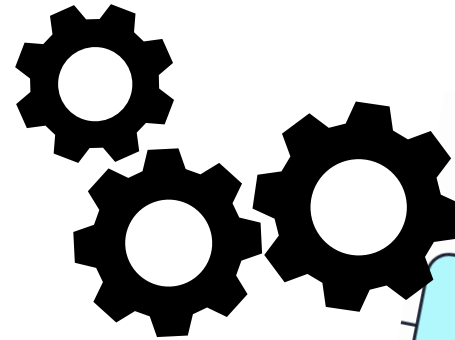
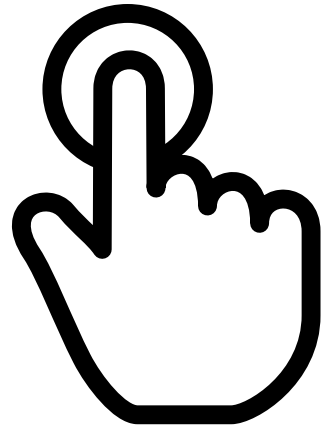
OPEN DATA

OPEN SOURCE

OPEN MATERIALS

PREREGISTERED

F_{indable} A_{ccessible} I_{nteroperable} R_{eusable}





Excel

Pros:

- Everyone has it
- Kind of easy to enter data?

Cons:

- Everyone has it
- Nothing is traceable
- Complete minefield
 - Sort
 - Filter
 - Separators
 - Language
 - Date and currency

Book1 - Excel

Niklas Edvall

File Home Insert Page Layout Formulas Data Review View Help Acrobat Tell me what you want to do Share

Clipboard Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat

E2

	A	B	C	D	E	F	G
1	ID	data1	data2				
2		1 Yes	21				
3		2 No	41				
4		3 Sometimes	23				
5		4 Maybe	64				
6		5 No	68				
7		6 Yes	23				
8		7 Often	53				
9		8 Never	64				
10							
11							
12							

Sheet1

Ready Accessibility: Good to go

205%

nature

NEWS | 13 August 2021 | Correction [25 August 2021](#)

Autocorrect errors in Excel still creating genomics headache

Despite geneticists being warned about spreadsheet problems, 30% of published papers contain mangled gene names in supplementary data.

Gene name errors are widespread in the scientific literature

[Mark Ziemann](#), [Yotam Eren](#) & [Assam El-Osta](#) 

[Genome Biology](#) **17**, Article number: 177 (2016) | [Cite this article](#)

139k Accesses | **67** Citations | **3223** Altmetric | [Metrics](#)

Abstract

The spreadsheet software Microsoft Excel, when used with default settings, is known to convert gene names to dates and floating-point numbers. A programmatic scan of leading genomics journals reveals that approximately one-fifth of papers with supplementary Excel gene lists contain erroneous gene name conversions.

B B C

 Sign in

Home

News

Sport

NEWS

Excel: Why using Microsoft's tool caused Covid-19 results to be lost

By [Leo Kelion](#)
Technology desk editor

 5 October 2020

CSV: Comma separated values

TSV: Tab separated values

```
"" , "age" , "sex" , "survey" , "TimeHere" , "position" , ...  
"1" , 49 , 1 , 4 , 6 , 4 , 3.4 , 7.8 , 4 , 4 , 5 , 4 , 3 , 0 , 2 , 2 , 1  
"2" , 50 , 2 , 2 , 10 , 3 , 4.5 , 7.4 , 3 , 4 , 2 , 3 , 4 , 1 , 2 , 1 , 1  
"3" , 47 , 1 , 3 , 3 , 4 , 5 , 8.3 , 3 , 4 , 3 , 1 , 3 , 3 , 1 , 1 , 0  
"4" , 61 , 1 , NA , 10 , 5 , 3.9 , 4 , 4 , 5 , 4 , 4 , 4 , 2 , 0 , 2 , 0  
"5" , 23 , 1 , 3 , 1 , 1 , 1.9 , 6.1 , 5 , 4 , 4 , 5 , 4 , 1 , 1 , 1 , 1  
"6" , 16 , 1 , 2 , 1 , 1 , 6.3 , 8.3 , 4 , 4 , 5 , 4 , 2 , 2 , 0 , 1 , 2  
"7" , 26 , 1 , 4 , 2 , 2 , 4.8 , 4 , 4 , 4 , 5 , 4 , 3 , 1 , 2 , 1 , 1
```

Find, share, reuse – your portal to research data

Type a keyword, dataset, or subject

 Search

Why should science be open?

Open science benefits research and society. It can also be of value for individual researchers.

[Read more >](#)

Join thousands of researchers – start sharing data

Researchdata.se is a simple gateway to several leading infrastructures for research data.

[Read more >](#)

Learn good research data management

We guide you through data management to give your data added value and a chance at eternal life.

[Read more >](#)

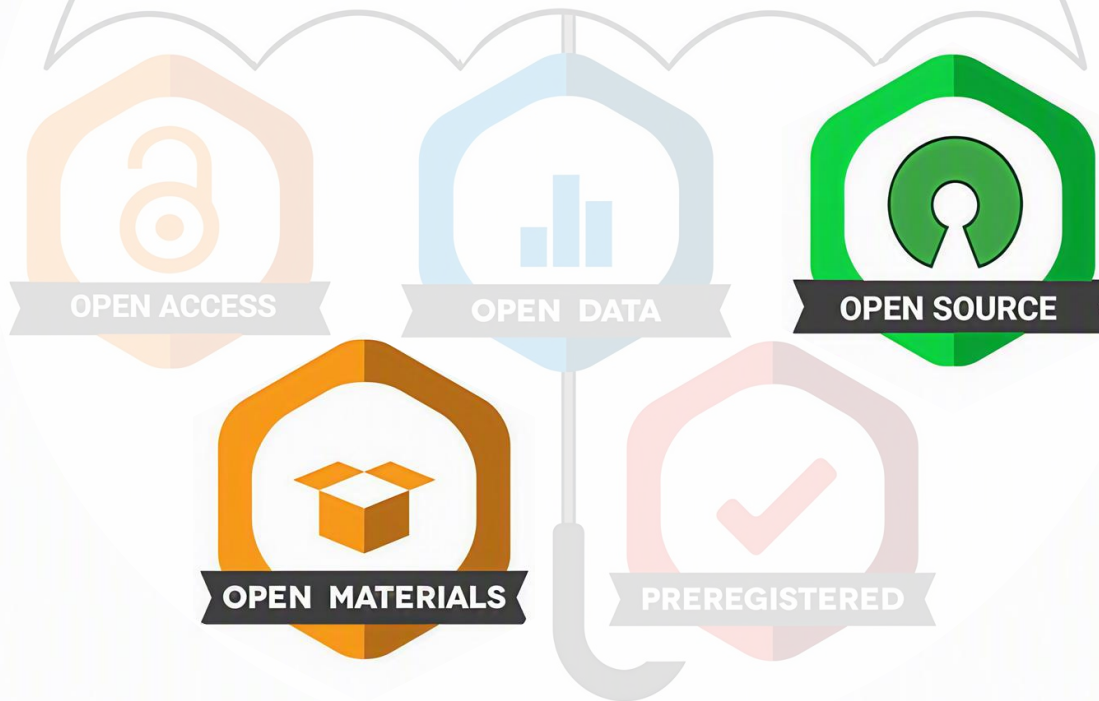
What is Researchdata.se?

Researchdata.se is a national web portal where you can find, share, and reuse research data from a wide range of research fields. You can also find helpful advice and recommendations on managing research data.

Several leading research infrastructures are behind Researchdata.se, which is an important step towards the goal of open access to publicly funded research data.

[Read more about Researchdata.se >](#)

OPEN SCIENCE



Article | Published: 20 May 2020

Variability in the analysis of a single neuroimaging dataset by many teams

[Rotem Botvinik-Nezer](#), [Felix Holzmeister](#), [Colin F. Camerer](#), [Anna Dreber](#), [Juergen Huber](#), [Magnus Johannesson](#), [Michael Kirchler](#), [Roni Iwanir](#), [Jeanette A. Mumford](#), [R. Alison Adcock](#), [Paolo Avesani](#), [Blazej M. Baczkowski](#), [Aahana Bajracharya](#), [Leah Bakst](#), [Sheryl Ball](#), [Marco Barilari](#), [Nadège Bault](#), [Derek Beaton](#), [Julia Beitner](#), [Roland G. Benoit](#), [Ruud M. W. J. Berkers](#), [Jamil P. Bhanji](#), [Bharat B. Biswal](#), [Sebastian Bobadilla-Suarez](#), ... [Tom Schonberg](#)  [+ Show authors](#)

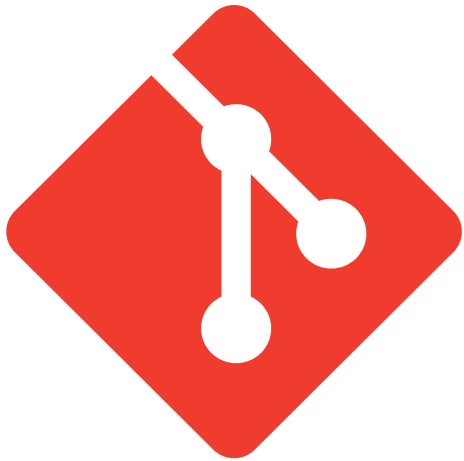
[Nature](#) **582**, 84–88 (2020) | [Cite this article](#)

69k Accesses | **959** Citations | **1873** Altmetric | [Metrics](#)

Abstract

Data analysis workflows in many scientific domains have become increasingly complex and flexible. Here we assess the effect of this flexibility on the results of functional magnetic resonance imaging by asking 70 independent teams to analyse the same dataset, testing the same 9 ex-ante hypotheses¹. The flexibility of analytical approaches is exemplified by the fact that no two teams chose identical workflows to analyse the data. This flexibility resulted in sizeable variation in the results of hypothesis tests, even for teams whose statistical maps were highly correlated at intermediate stages of the analysis pipeline. Variation in reported results was related to several aspects of analysis methodology. Notably, a meta-analytical approach that aggregated information across teams yielded a significant consensus in activated regions. Furthermore, prediction markets of researchers in the field revealed an overestimation of the likelihood of significant findings, even by researchers with direct knowledge of the dataset^{2,3,4,5}. Our findings show that analytical flexibility can have substantial effects on scientific conclusions, and identify factors that may be related to variability in the analysis of functional magnetic resonance imaging. The results emphasize the importance of validating and sharing complex analysis workflows, and demonstrate the need for performing and reporting multiple analyses of the same data. Potential approaches that could be used to mitigate issues related to analytical variability are discussed.

Version control

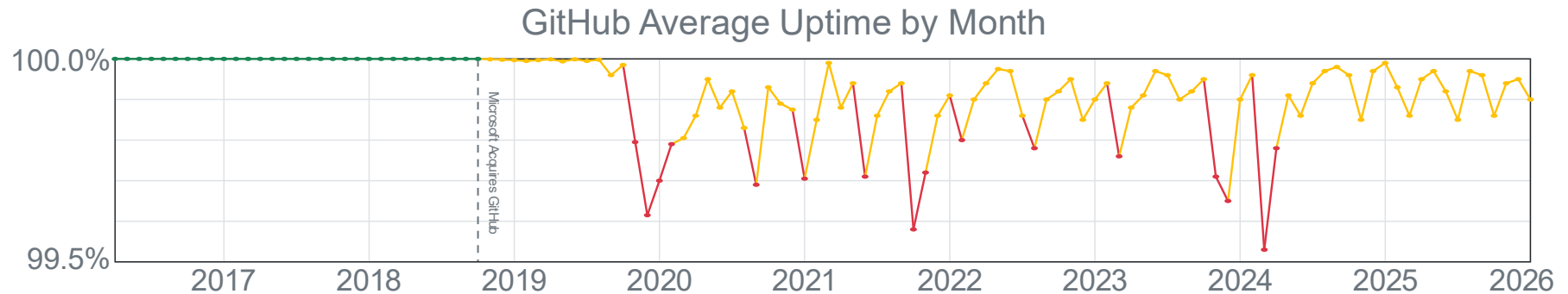


git



<https://github.com/jj-vcs/jj>

Version control – remote hosts



Paus?



important-project/

data/



raw



processed



figures

analysis/



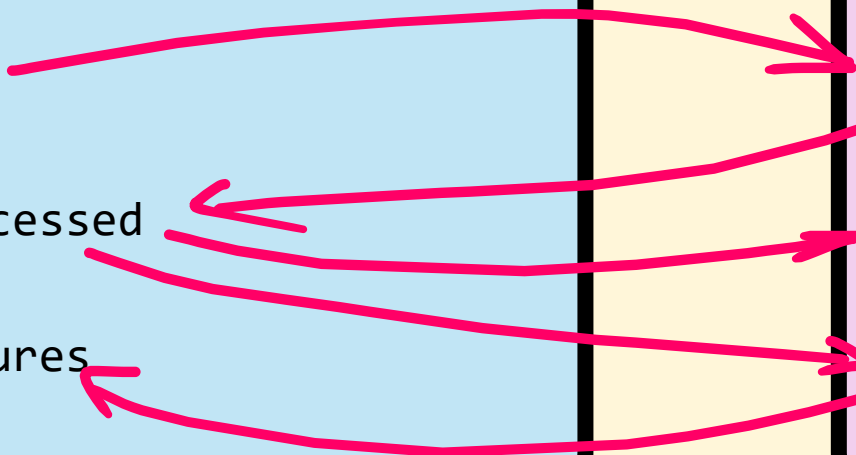
clean-raw.r



linear-model.r



figure1.r



important-project/

data/



raw



processed



figures

analysis/



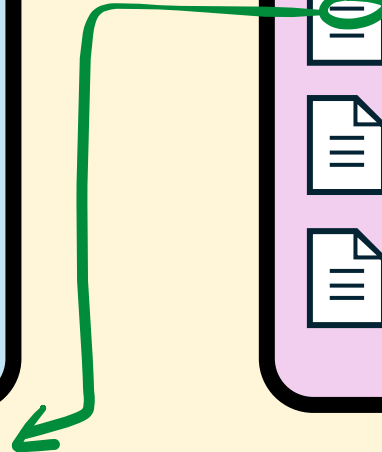
clean-raw.r



linear-model.r

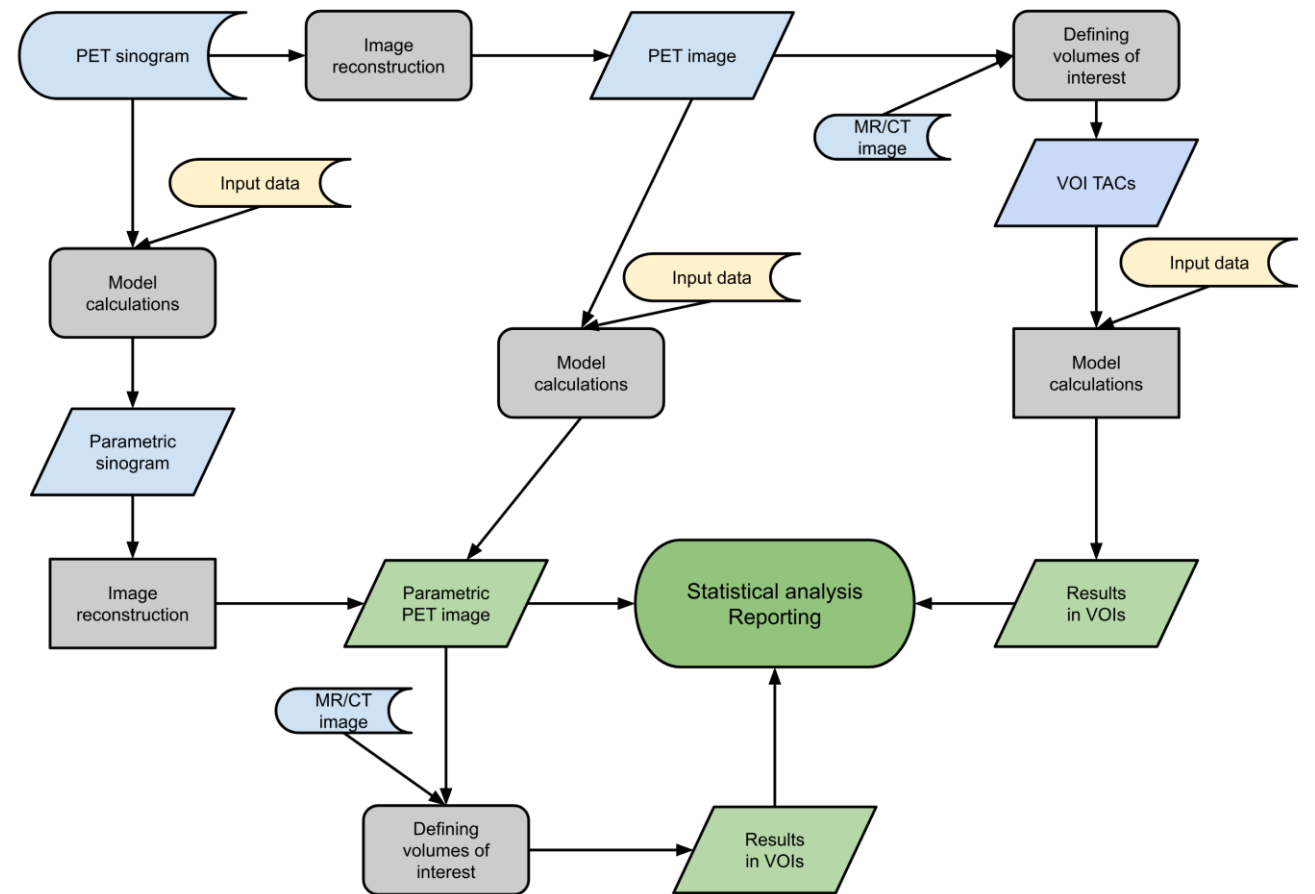
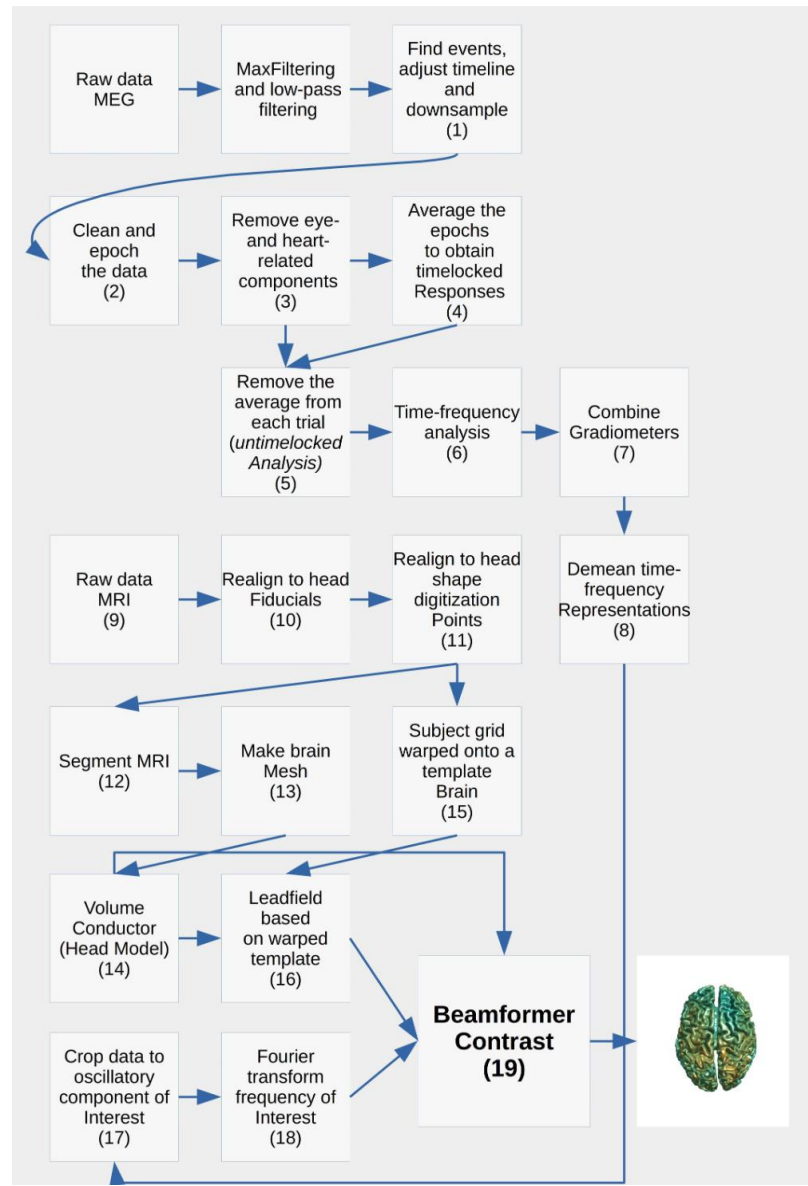


figure1.r

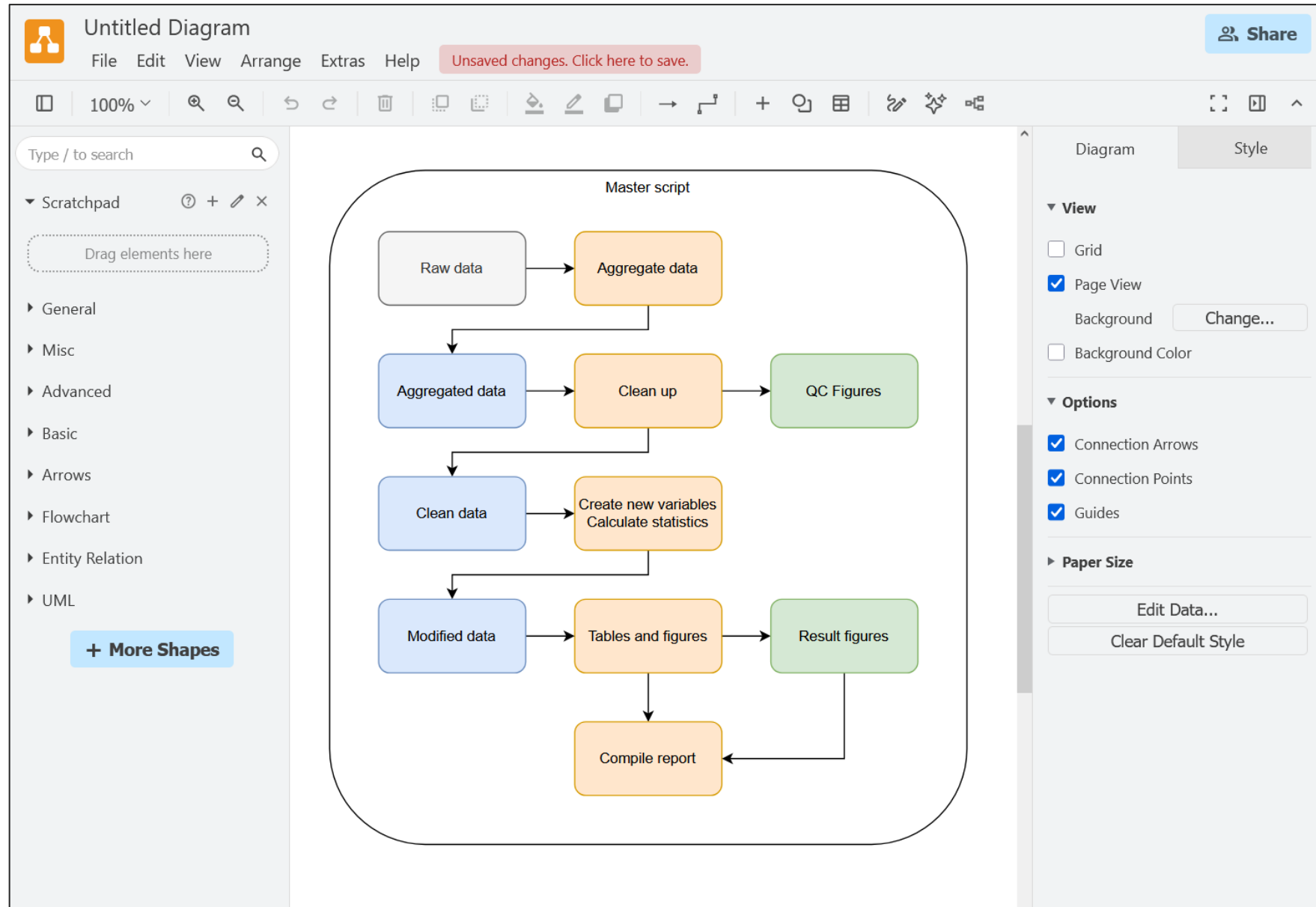


Absolute: `read.csv("C:/research/nikedv/important-project/data/raw/data-file.csv")`

Relative: `read.csv("../data/raw/data-file.csv")`



draw.io
<https://app.diagrams.net/>



README.md – Code is NOT documentation

The screenshot shows a GitHub repository named "an-approved-repo" by user "NiklasEdvall". The repository is public and has 1 branch and 0 tags. The main branch is selected. The repository contains the following files:

File	Description	Last Commit
data	simulated data and code key	4 hours ago
scripts	added some descriptive text to report template	10 minutes ago
.gitignore	gitignore	1 hour ago
README.md	Revise README for clarity and organization	4 minutes ago
run_pipeline.R	master script to run pipeline	1 hour ago

The README.md file is open, showing the following content:

an-approved-repo

This is an example of an organized repo for the course [Open Science in Practice: Collaborative Research with Git](#).

Purpose

- Create a simulated dataset of 150 patients with either no, mild or severe tinnitus, some basic demographics and results from a hearing test (pure tone audiometry + UCL).
- Clean up, visualize and summarize the simulated data in an organized and well described pipeline.
- Use quarto markdown template to generate a summary report as html

Dependencies

The repository also includes a sidebar with the following information:

- About:** This is supposed to be a good example of a well organized repo for the course "Open Science in Practice: Collaborative Research with Git"
- Readme:** Link to the README file
- Activity:** Link to the repository activity
- Stars:** 0 stars
- Watching:** 0 watching
- Forks:** 0 forks
- Releases:** No releases published. [Create a new release](#)
- Packages:** No packages published. [Publish your first package](#)
- Contributors:** 1 contributor: NiklasEdvall
- Languages:** R 100.0%

README.md – Exact instructions challenge



README.md – Exact instructions challenge



EDUCATION

Seven quick tips for analysis scripts in neuroimaging

Marijn van Vliet  *

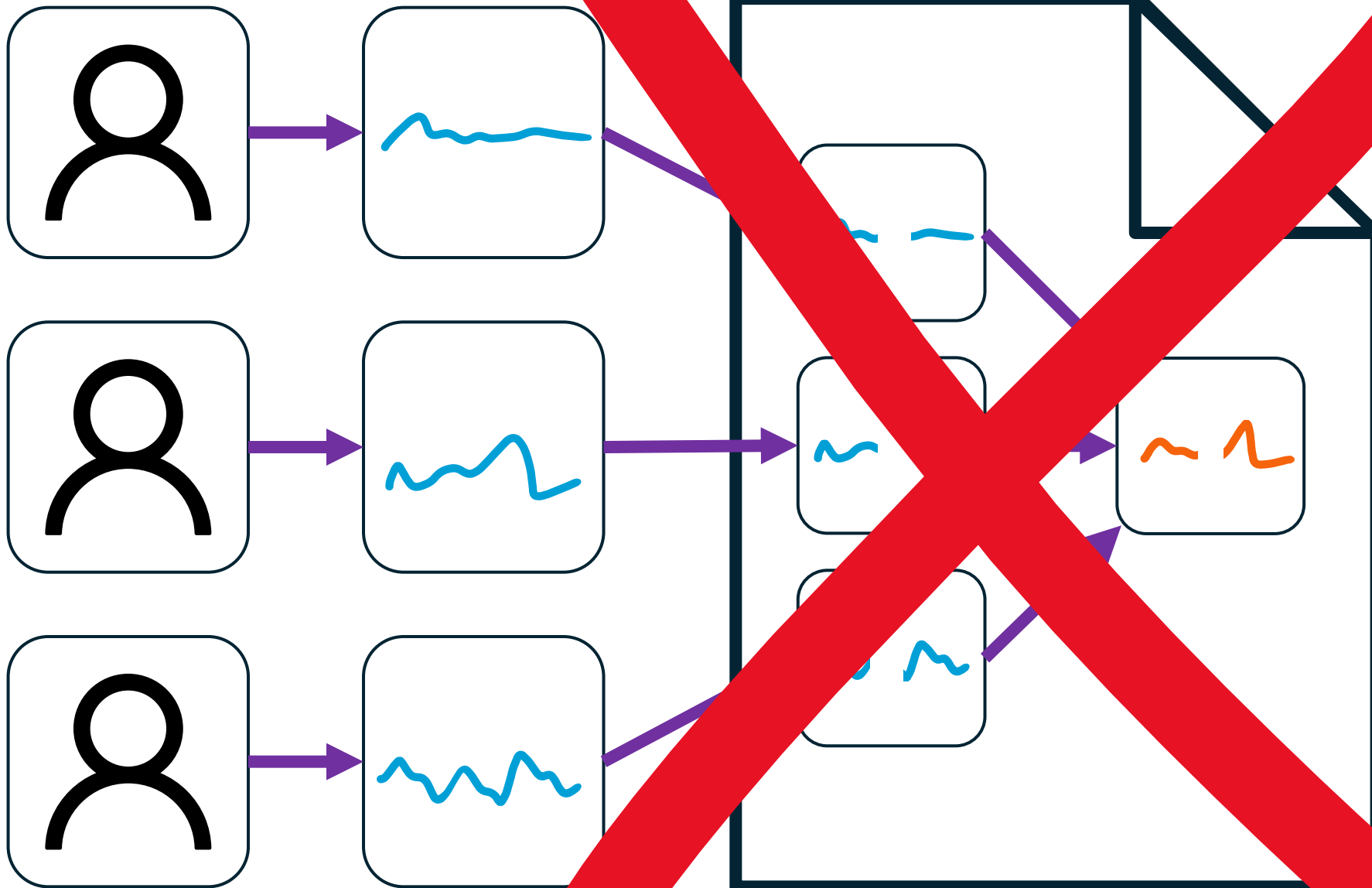
Department of Neuroscience and Biomedical Engineering, Aalto University, Espoo, Finland

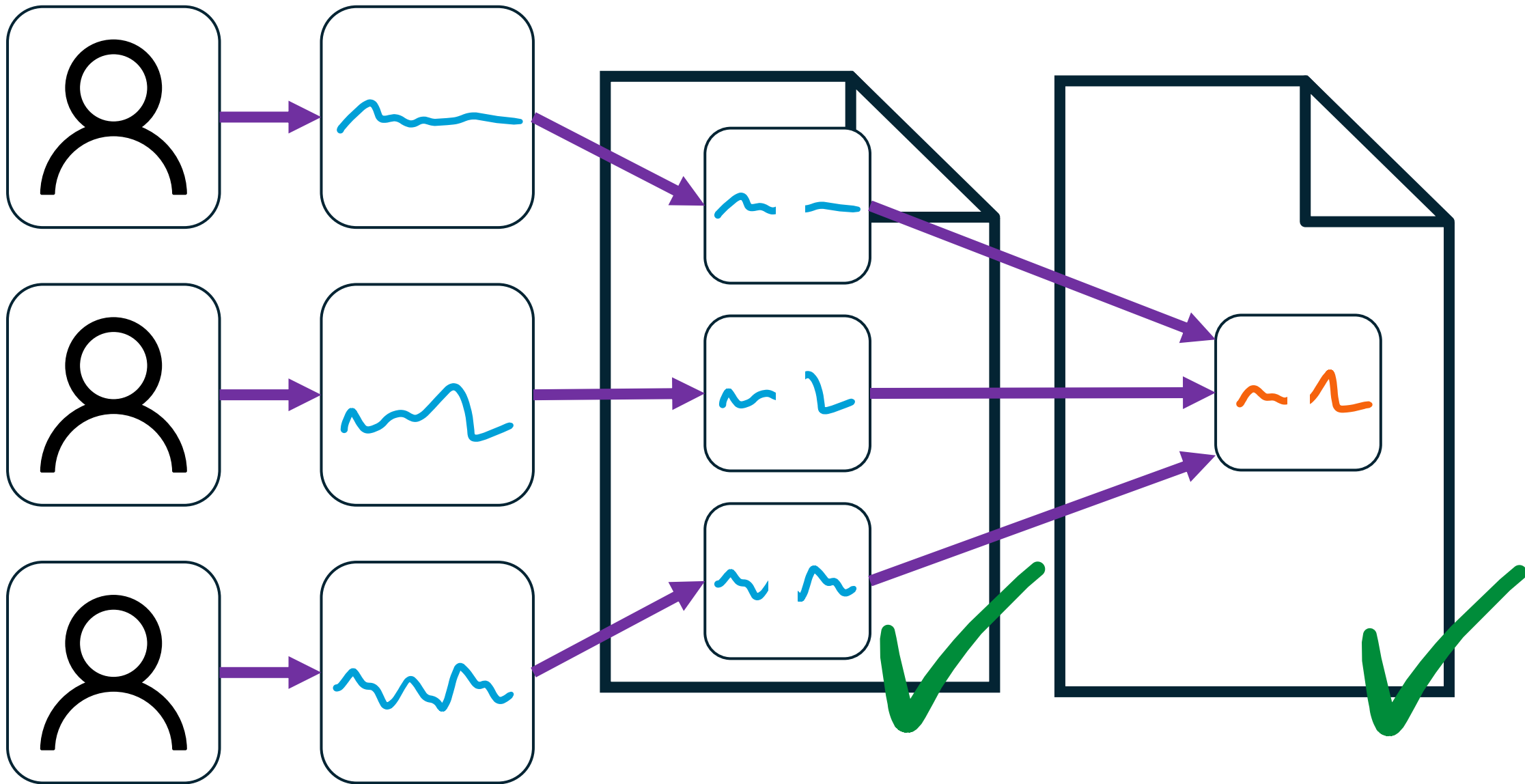
* marijn.vanvliet@aalto.fi

Introduction

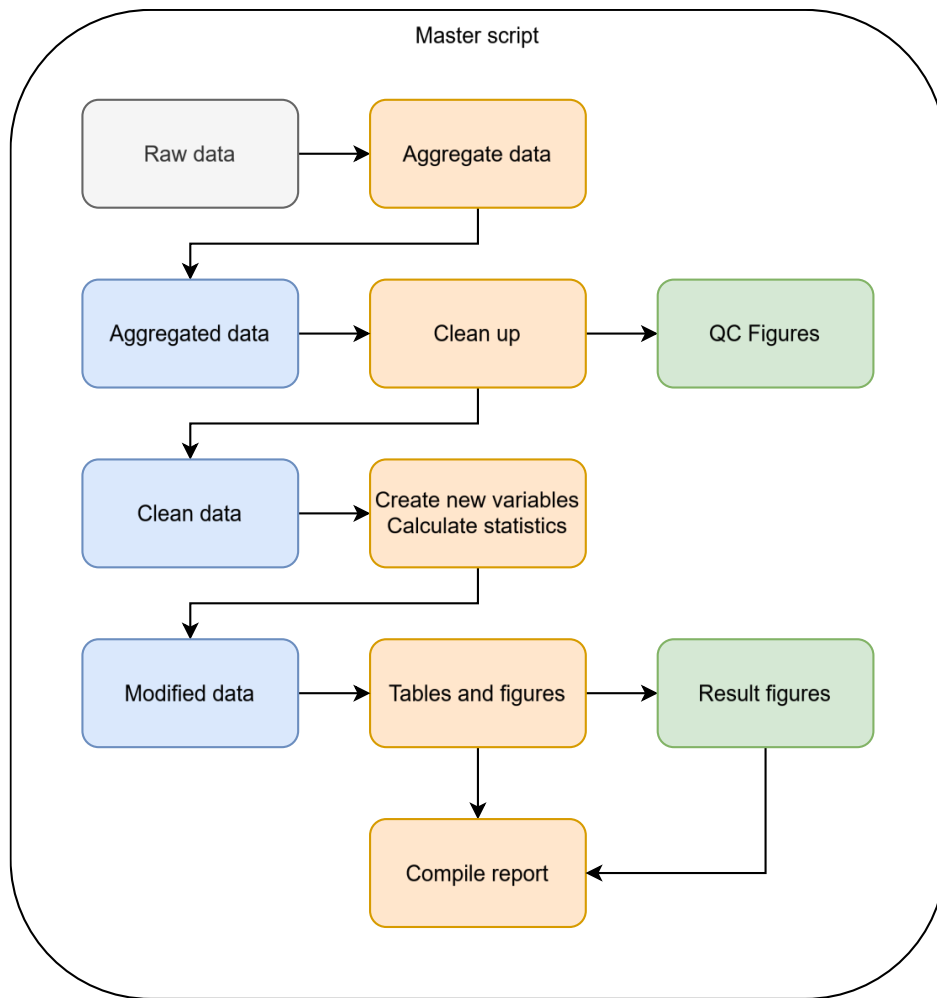
In neuroimaging, as in many fields, the journey of the data from the measurement equipment to a figure in a publication is growing longer and more complicated. New preprocessing steps are added at the beginning of the pipeline [1–3], new multivariate techniques in the middle

1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both

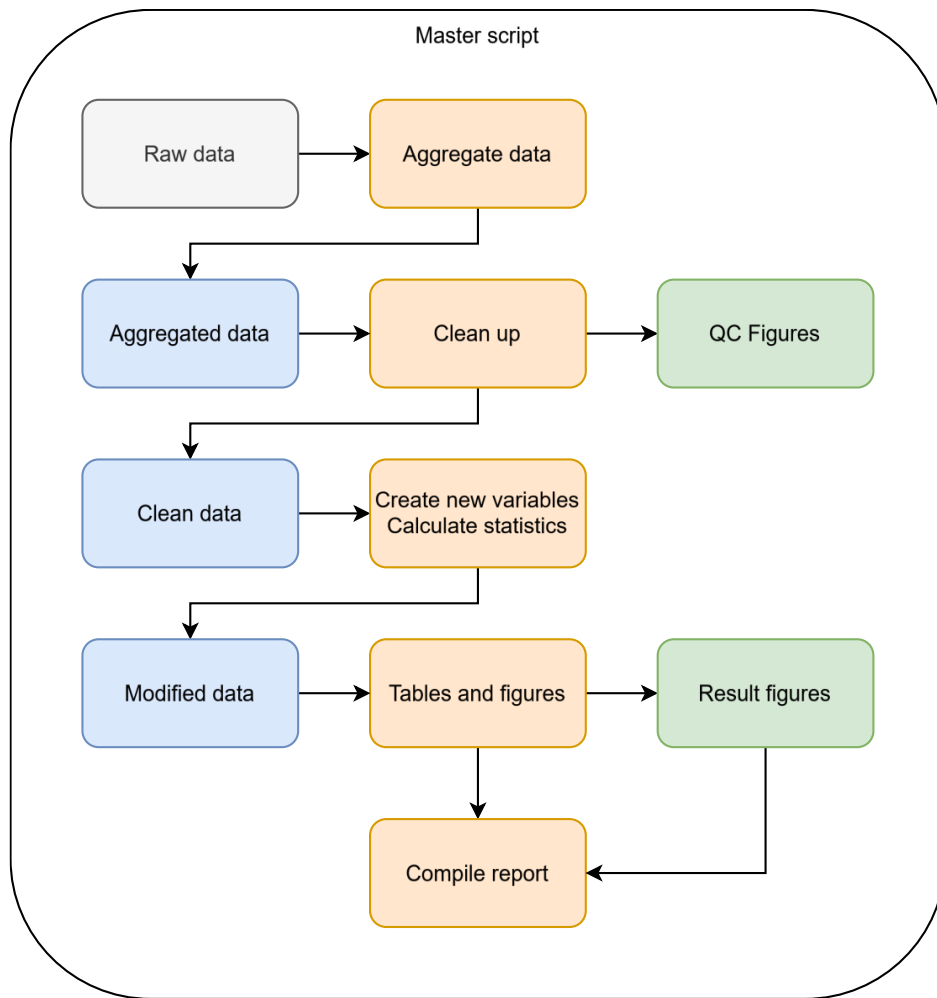




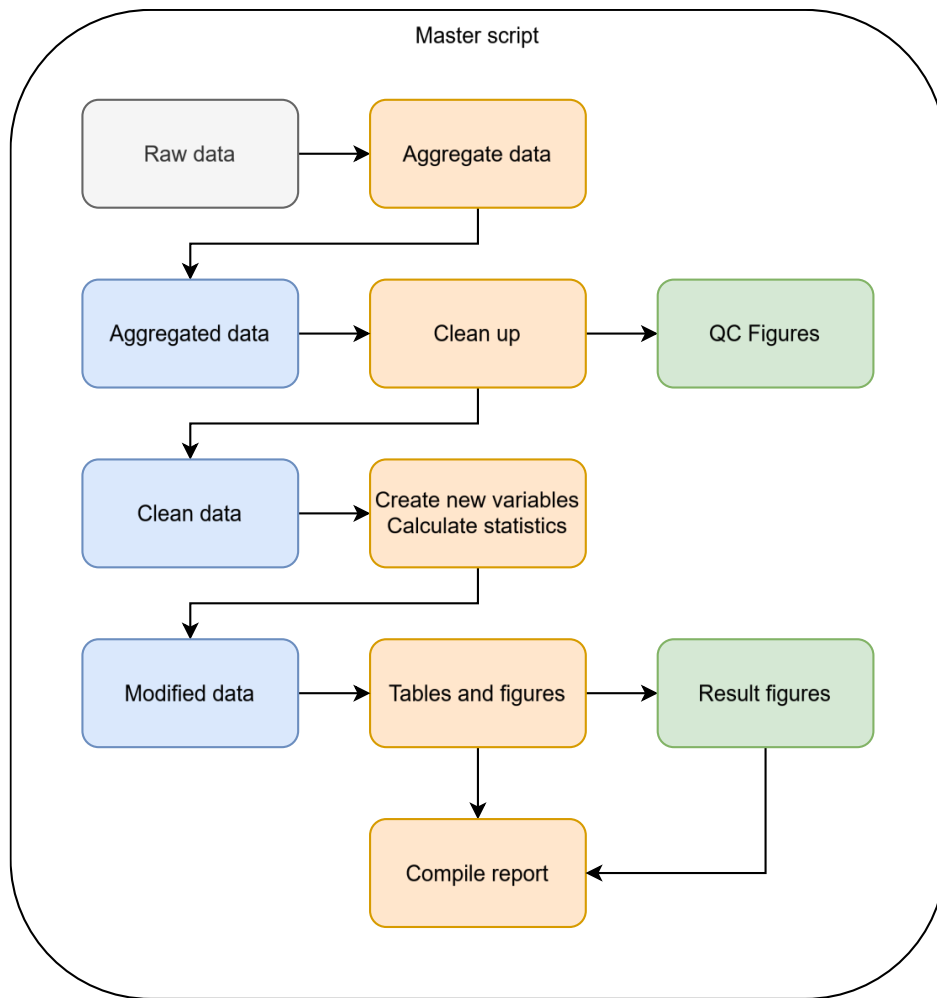
1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both
3. One master script to run the entire analysis



```
master.r U
master.r > ...
1
2 # Project and data locations
3 DATA_ROOT <- "../data"
4 OUTPUT_ROOT <- "../results"
5
6 # Input raw data
7 DATA_FILE <- file.path(DATA_ROOT, "data.csv")
8
9 # Basic analysis parameters
10 PARAMS <- list(
11   alpha = 0.05,
12   na_action = "exclude",
13   qc = TRUE
14 )
15
16 # Create output directories if needed
17 dir.create(OUTPUT_ROOT, showWarnings = FALSE)
18 dir.create(file.path(OUTPUT_ROOT, "qc"), showWarnings = FALSE)
19
20 # Load data
21 data_raw <- read.csv(DATA_FILE)
22
23 # Run script to aggregate data
24 source("scripts/01_aggregate_data.R")
25
26 # Clean aggregated data
27 source("scripts/02_clean_data.R")
28
29 # Quality control
30 source("scripts/03_qc_figures.R")
31
32 # Create variables and calculate statistics
33 source("scripts/04_analysis.R")
34
```



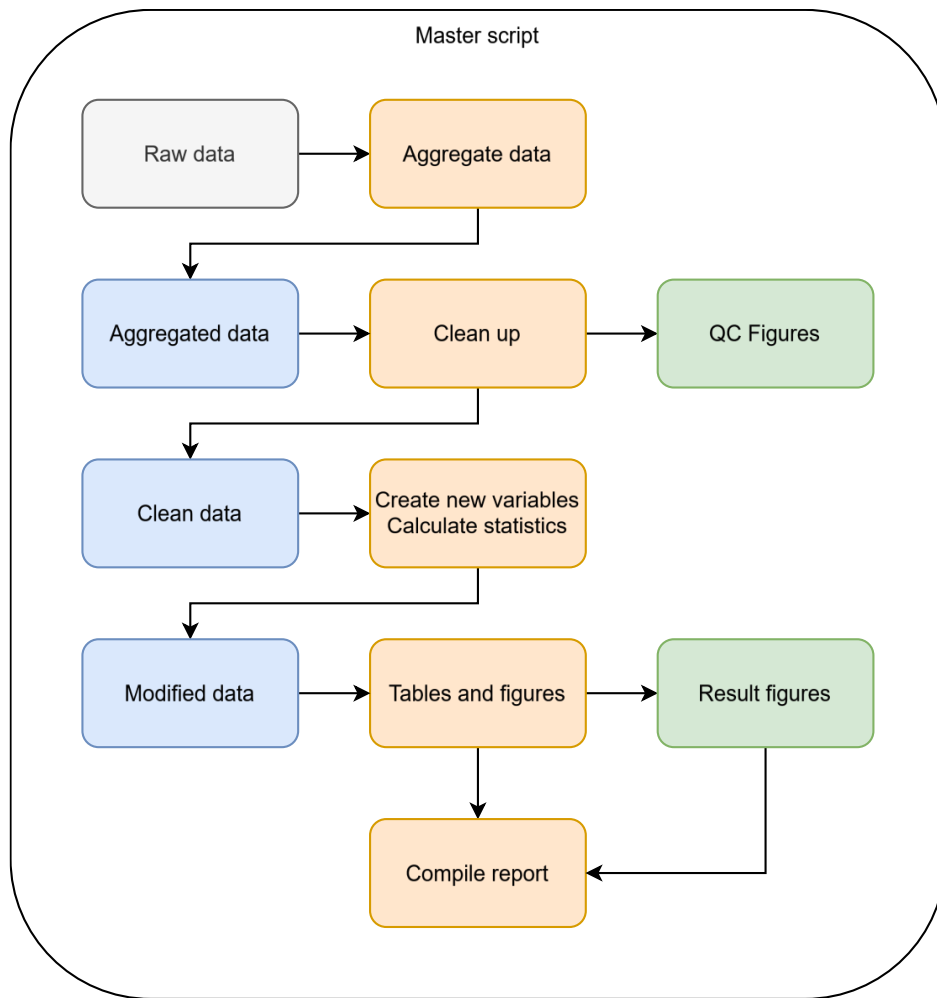
```
master.r U
master.r > ...
1
2 # Project and data locations
3 DATA_ROOT <- "../data"
4 OUTPUT_ROOT <- "../results"
5
6 # Input raw data
7 DATA_FILE <- file.path(DATA_ROOT, "data.csv")
8
9 # Basic analysis parameters
10 PARAMS <- list(
11   alpha = 0.05,
12   na_action = "exclude",
13   qc = TRUE
14 )
15
16 # Create output directories if needed
17 dir.create(OUTPUT_ROOT, showWarnings = FALSE)
18 dir.create(file.path(OUTPUT_ROOT, "qc"), showWarnings = FALSE)
19
20 # Load data
21 data_raw <- read.csv(DATA_FILE)
22
23 # Run script to aggregate data
24 source("scripts/01_aggregate_data.R")
25
26 # Clean aggregated data
27 source("scripts/02_clean_data.R")
28
29 # Quality control
30 source("scripts/03_qc_figures.R")
31
32 # Create variables and calculate statistics
33 source("scripts/04_analysis.R")
34
```



```
master.r U
master.r > ...
1
2 # Project and data locations
3 DATA_ROOT <- "../data"
4 OUTPUT_ROOT <- "../results"
5
6 # Input raw data
7 DATA_FILE <- file.path(DATA_ROOT, "data.csv")
8
9 # Basic analysis parameters
10 PARAMS <- list(
11   alpha = 0.05,
12   na_action = "exclude",
13   qc = TRUE
14 )
15
16 # Create output directories if needed
17 dir.create(OUTPUT_ROOT, showWarnings = FALSE)
18 dir.create(file.path(OUTPUT_ROOT, "qc"), showWarnings = FALSE)
19
20 # Load data
21 data_raw <- read.csv(DATA_FILE)
22
23 # Run script to aggregate data
24 source("scripts/01_aggregate_data.R")
25
26 # Clean aggregated data
27 source("scripts/02_clean_data.R")
28
29 # Quality control
30 source("scripts/03_qc_figures.R")
31
32 # Create variables and calculate statistics
33 source("scripts/04_analysis.R")
34
```

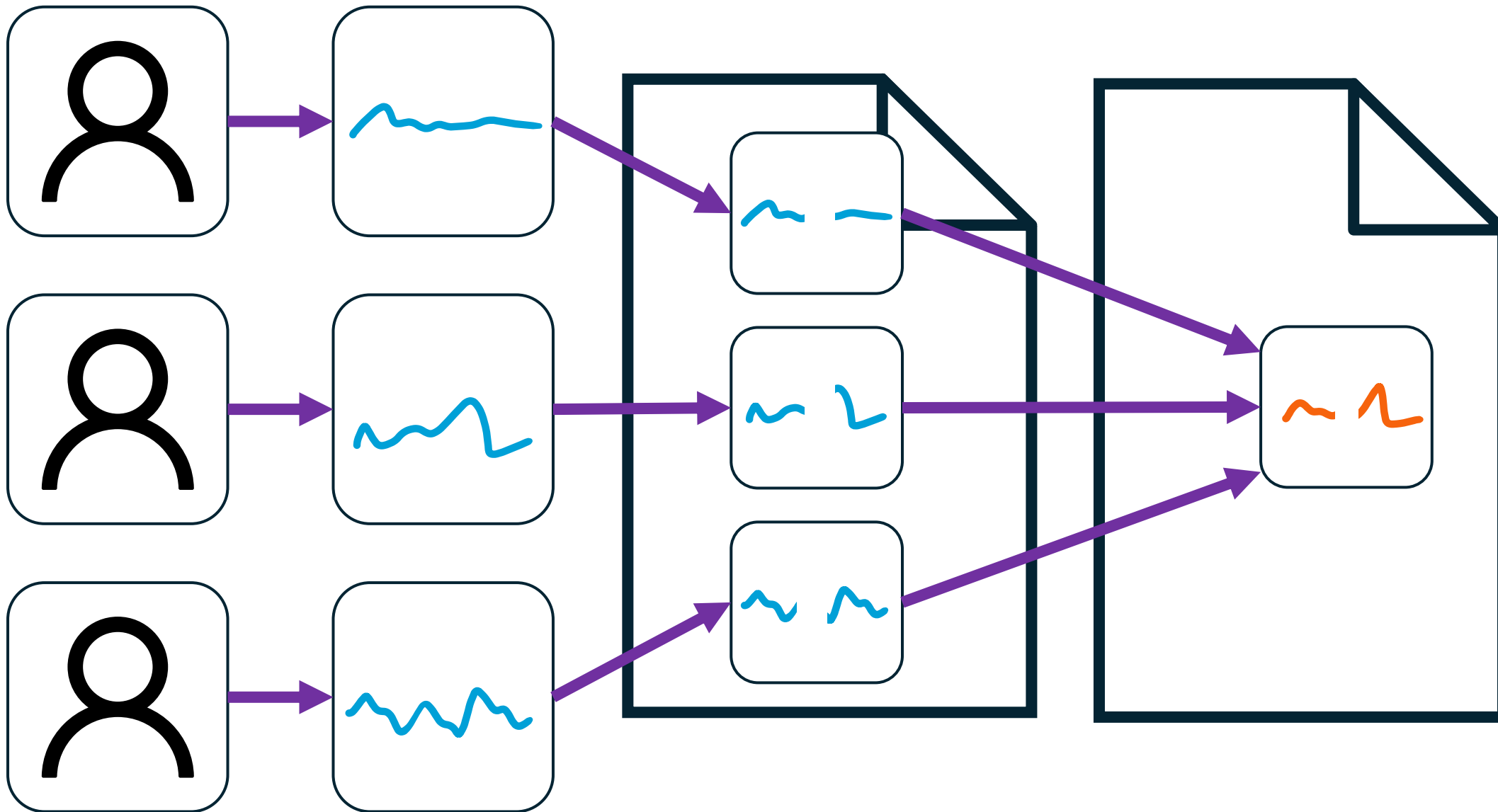
1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both
3. One master script to run the entire analysis

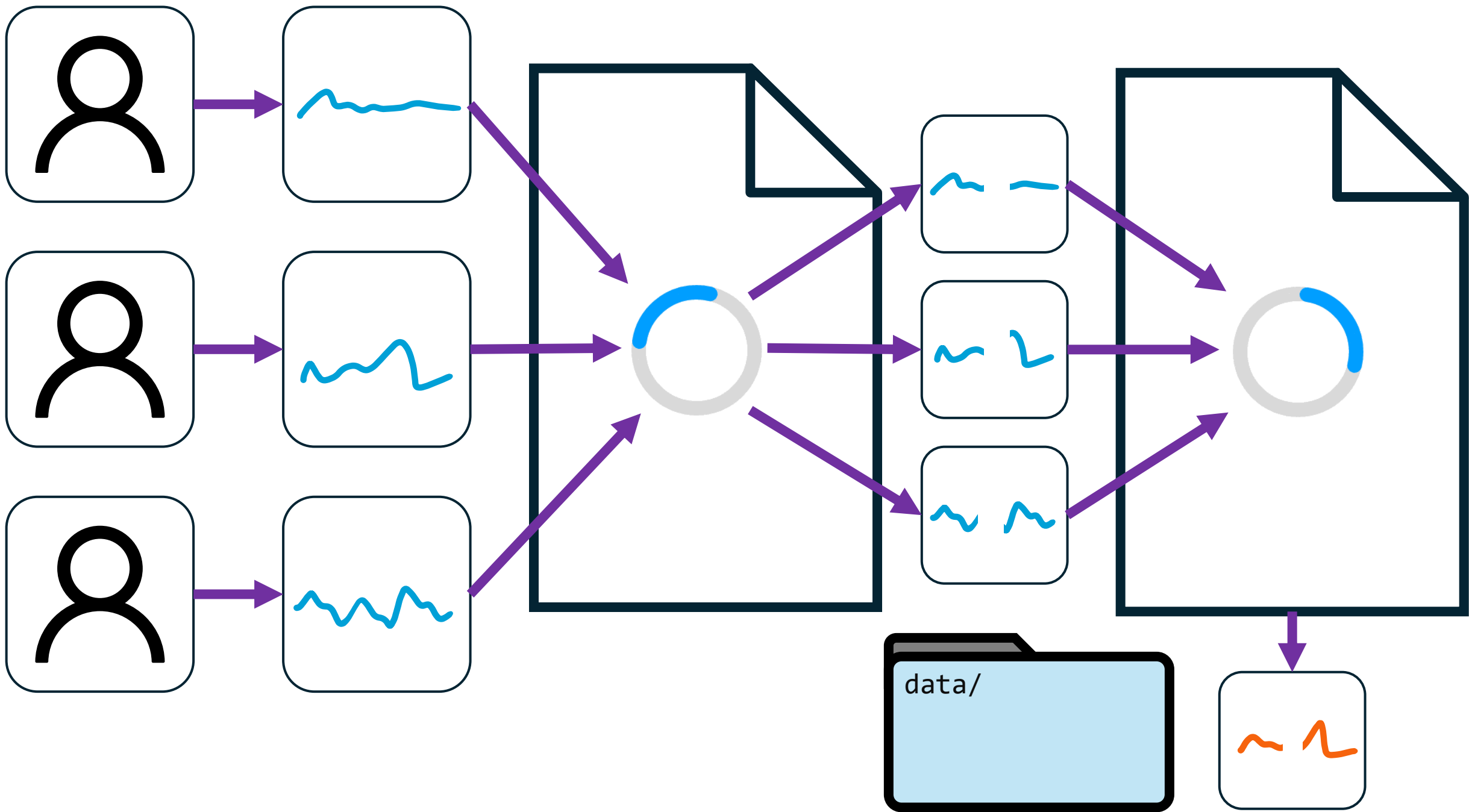
6. Each parameter and file name is defined only once

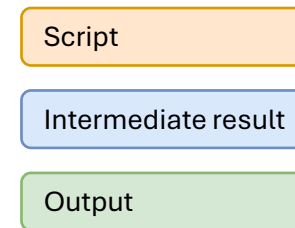
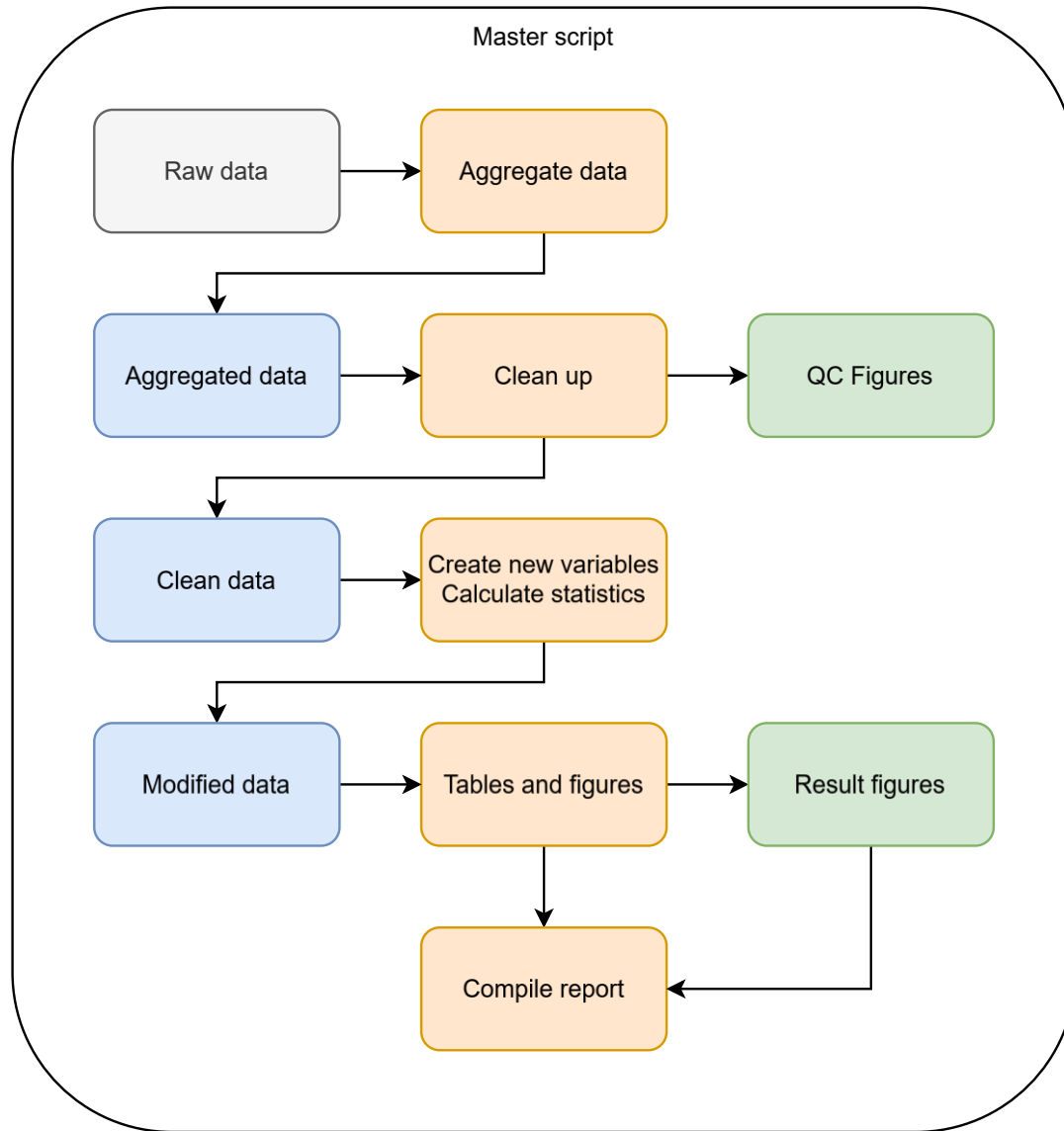


```
master.r U
master.r > ...
1
2 # Project and data locations
3 DATA_ROOT <- "../data"
4 OUTPUT_ROOT <- "../results"
5
6 # Input raw data
7 DATA_FILE <- file.path(DATA_ROOT, "data.csv")
8
9 # Basic analysis parameters
10 PARAMS <- list(
11   alpha = 0.05,
12   na_action = "exclude",
13   qc = TRUE
14 )
15
16 # Create output directories if needed
17 dir.create(OUTPUT_ROOT, showWarnings = FALSE)
18 dir.create(file.path(OUTPUT_ROOT, "qc"), showWarnings = FALSE)
19
20 # Load data
21 data_raw <- read.csv(DATA_FILE)
22
23 # Run script to aggregate data
24 source("scripts/01_aggregate_data.R")
25
26 # Clean aggregated data
27 source("scripts/02_clean_data.R")
28
29 # Quality control
30 source("scripts/03_qc_figures.R")
31
32 # Create variables and calculate statistics
33 source("scripts/04_analysis.R")
34
```

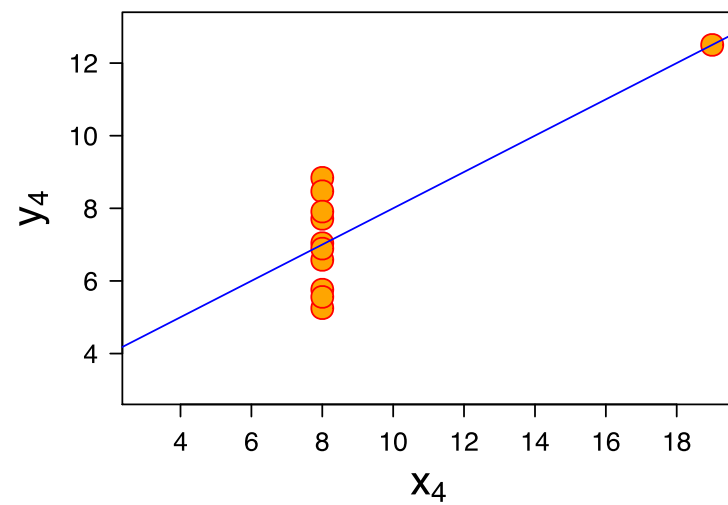
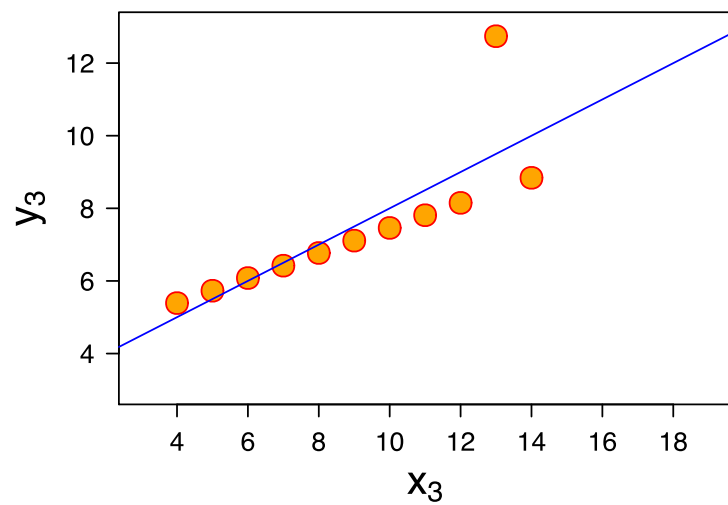
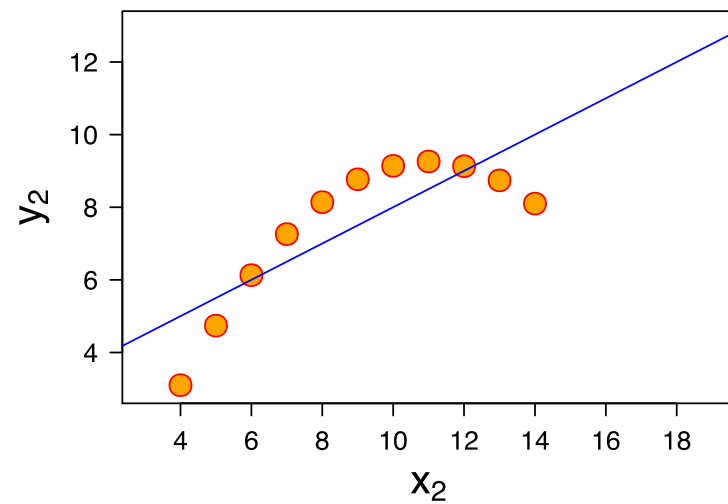
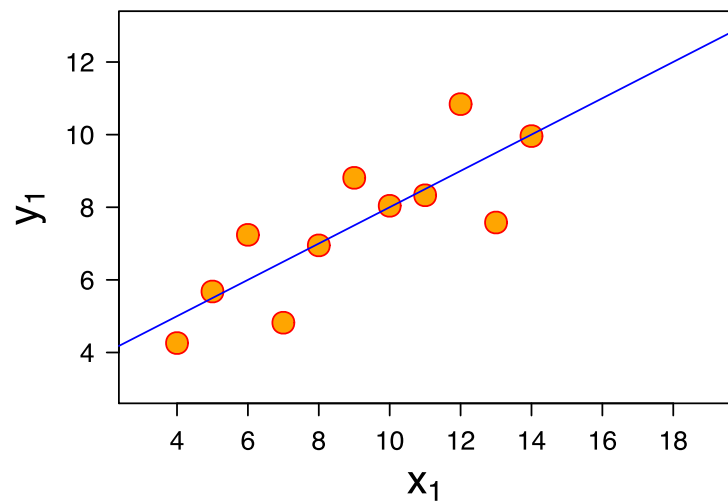
1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both
3. One master script to run the entire analysis
4. Save all intermediate results
6. Each parameter and file name is defined only once

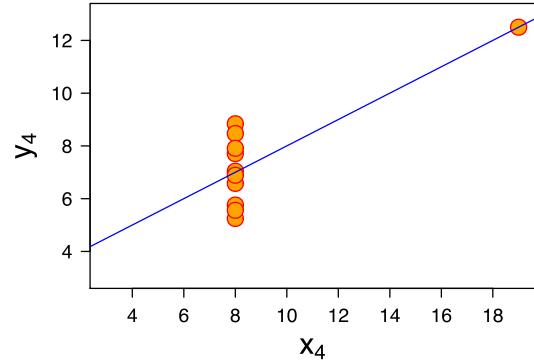
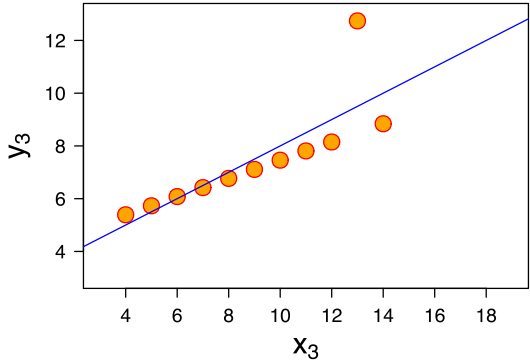
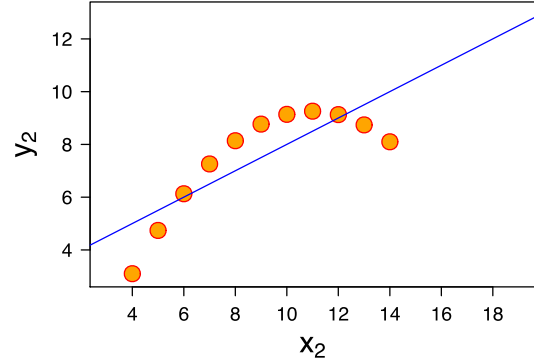
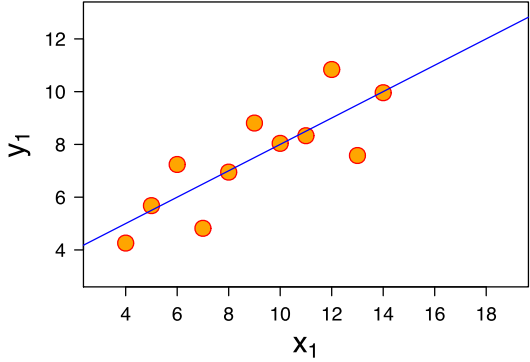






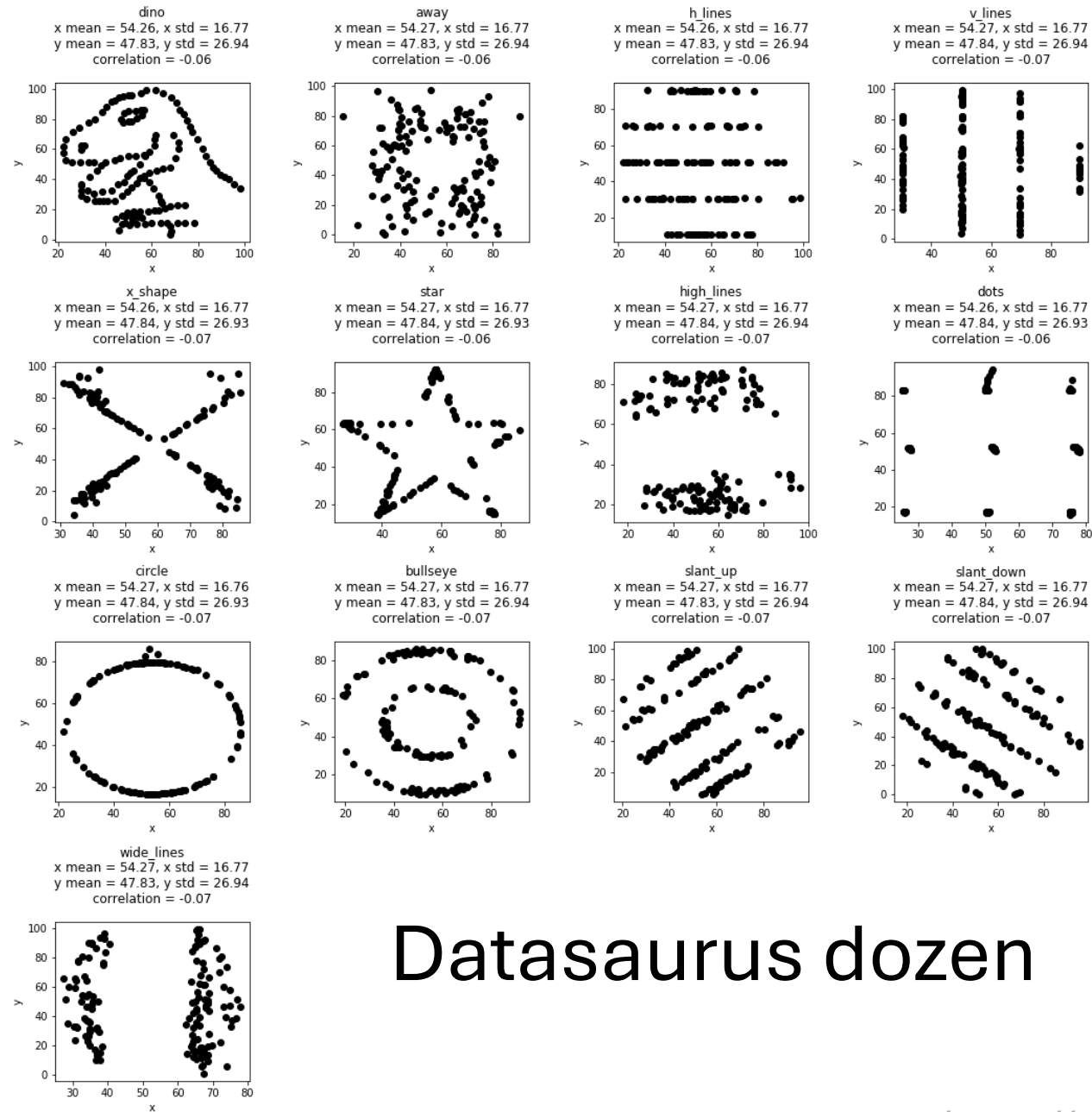
1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both
3. One master script to run the entire analysis
4. Save all intermediate results
5. Visualize all intermediate results
6. Each parameter and file name is defined only once





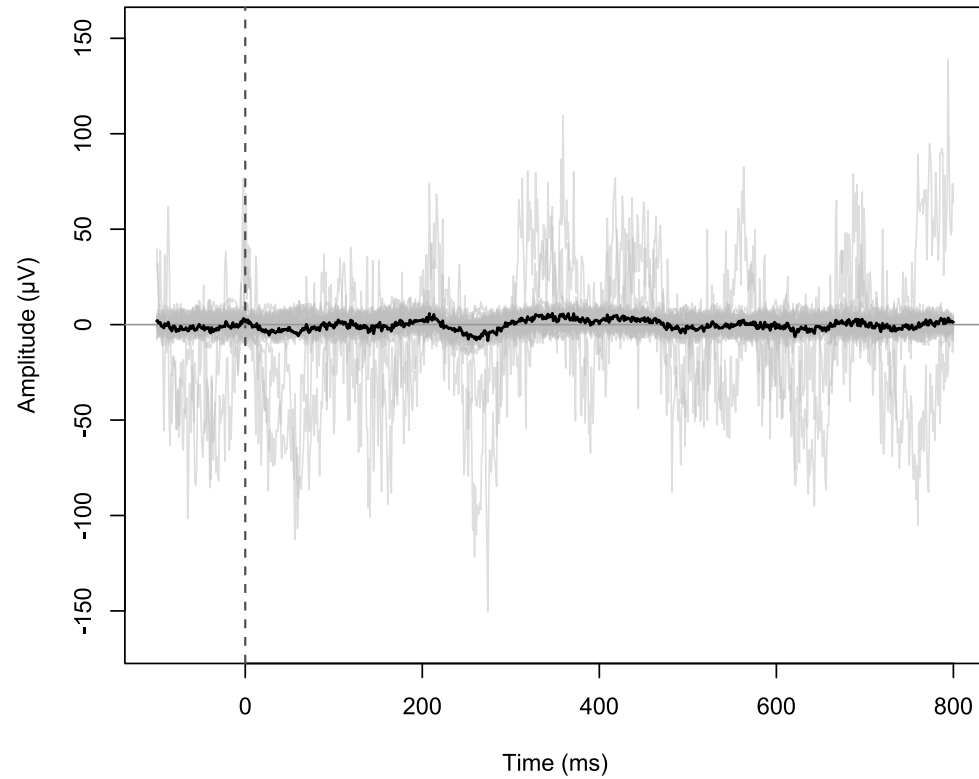
Anscombe's quartet

Property	Value	Accuracy
Mean of x	9	exact
Sample variance of x	11	exact
Mean of y	7.50	to 2 decimal places
Sample variance of y	4.125	± 0.003
Correlation between x and y	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places
Coefficient of determination of the linear regression: R^2	0.67	to 2 decimal places

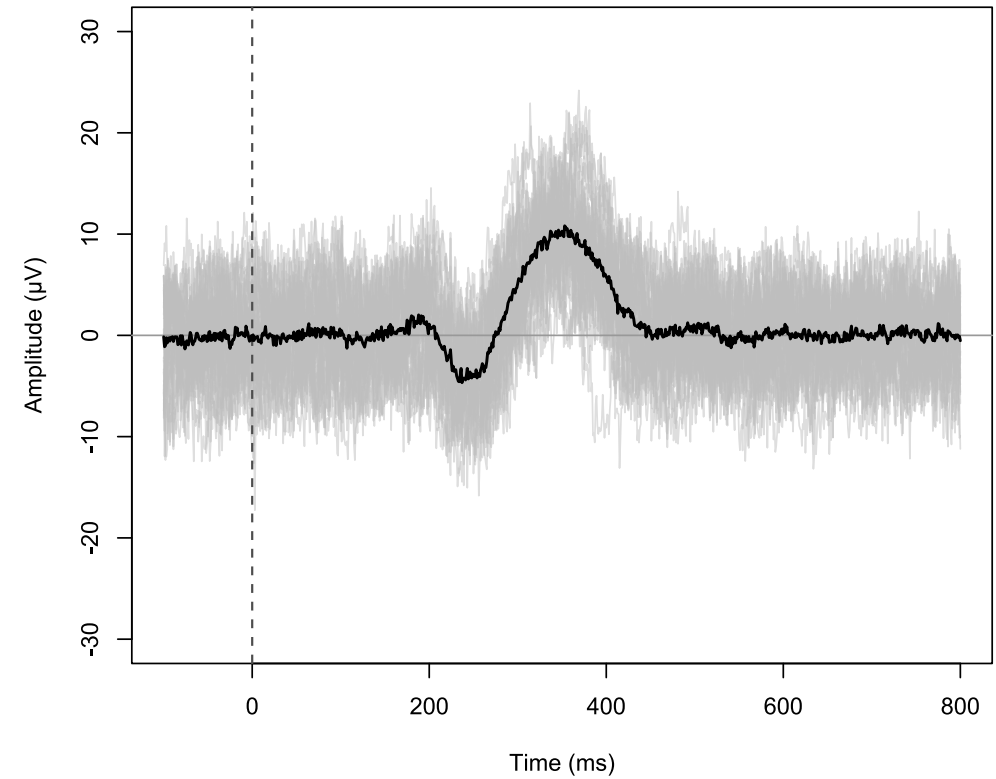


Datasaurus dozen

EEG trials with artefacts



Trials excluded



1. Each analysis step is one script
2. A script either processes a single recording or aggregates across recordings, never both
3. One master script to run the entire analysis
4. Save all intermediate results
5. Visualize all intermediate results
6. Each parameter and file name is defined only once
7. Distinguish files that are part of the official pipeline from other scripts

Remember

- The value of Open science (Data & Materials)
- Reproducibility lives in code
- Never store data in excel
- Analysis not on github doesn't exist
- Keep an updated overview (flowchart) of your pipeline
- Code is not documentation
- The seven quick tips

Open Science in Practice: Collaborative Research with Git



Niklas Edvall, PhD
niklas.edvall@ki.se

K8F6106
HT2026



Andreas Gerhardsson, PhD
andreas.Gerhardsson@ki.se